ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Principle of Communication

English Medium

EXERCISE-O

SINGLE CORRECT TYPE QUESTIONS

1. If a carrier wave of 1000 kHz is used to carry the signal, the length of transmitting antenna will be equal to :-

(A) 3 m (B) 30m (C) 300 m (D) 3000 m

PC0005

2. The maximum range of ground or surface wave propagation depends on :-

(A) the frequency of the radio waves only (B) power of the transmitter only

(C) both of them (D) none of them

PC0006

3. For television broadcasting, the frequency employed is normally :-

(A) 30-300MHz (B) 30-300 GHz (C) 30-300KHz (D) 30-3OOHz-

PC0007

4. The audio signal :-

(A) can be sent directly over the air for large distance

(B) cannot be sent directly over the air for large distance

(C) possess very high frequency

(D) none of the above

PC0008

5. For a carrier frequency of 100 kHz and a modulating frequency of 5 kHz what is the width of AM transmission-

(A) 5 kHz (B) 10kHz (C) 20 kHz (D) 200 KHz

PC0009

6. Three waves A, B and C of frequencies 1600 kHz, 5 MHz and 60 MHz, respectively are to be transmitted from one place to another. Which of the following is the most appropriate mode of communication :

(A) A is transmitted via space wave while B and C are transmitted via sky wave.

(B) A is transmitted via ground wave, B via sky wave and C via space wave.

(C) B and C are transmitted via ground wave while A is transmitted via sky wave.

(D) B is transmitted via ground wave we while A and C are transmitted via space wave.

PC0010

7. A speech signal of 3kHz is used to modulate a carrier signal of frequency 1 MHz, using amplitude modulation. The frequencies of the side bands will be

(A) 1.003 MHz and 0.997 MHz. (B) 3001 kHz and 2997 kHz.

(C) 1003 kHz and 1000 kHz. (D) 1 MHz and 0.997 MHz.

PC0011

8. A message signal of frequency ω_m is superposed on a carrier wave of frequency ω_e to get an amplitude mediated wave (AM) The frequency of the AM wave will be

(A) ω_m (B) ω_e (C) $\frac{\omega_e + \omega_m}{2}$ (D) $\frac{\omega_e - \omega_m}{2}$

PC0012

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9. A basic communication system consists of
(A) transmitter (B) information source. (C) user of information.
(D) channel (E) receiver.
Choose the correct sequence in which these are arranged in a basic communication system :-
(A) ABCDE (B) BADEC (C) BDACE (D) BEADC

PC0013

10. **Statement 1 :** Skywave can not be observed on moon.
Statement 2 : Atmosphere of variable refractive index is require for propagation of skywave.
(A) Both Statement-1 and Statement-2 are true, and Statement-2 is the correct explanation of Statement-1
(B) Both Statement-1 and Statement-2 are true but Statement-2 is not the correct explanation of Statement- 1.
(C) Statement-1 is true but Statement-2 is false.
(D) Statement-1 is false but Statement-2 is true.

PC0014

11. **Statement 1 :** Ground wave communication is effective only allow frequencies in the range 500kHz to about 1500 kHz.
Statement 2 : The decrease in the intensity of the signal due to absorption by the earth and its atmosphere is higher for higher frequencies.
(A) Both Statement-1 and Statement-2 are true, and Statement-2 is the correct explanation of Statement-1.
(B) Both Statement-1 and Statement-2 are true but Statement-2 is not the correct explanation of Statement-I .
(C) Statement-1 is true but Statement-2 is false.
(D) Statement-1 is false but Statement-2 is true.

PC0015

12. A radar has a power of 1 kW and is operating at a frequency of 10 GHz. It is located on a mountain top of height 500 m. The maximum distance up to which it can detect object located on the surface of the earth (Radius of earth = 6.4×10^6 m) is : **[AIEEE-2012]**
(A) 40 km (B) 64 (C) 80 km (D) 16km

PC0019

MULTIPLE CORRECT TYPE QUESTIONS

13. An audio signal of I5kHz frequency cannot be transmitted over long distances without modulation because
(A) the size of the required antenna would be at least 5 km which is not convenient.
(B) the audio signal cannot be transmitted through skywaves.
(C) the size of the required antenna would be at least 20 km, which is not convenient.
(D) effective power transmitted would be very low, if the size of the antenna is less than 5 km.

PC0016

14. Audio sine waves of 3 kHz frequency are used to amplitude modulate a carrier signal of 1.5 MHz. Which of the following statements are true?
(A) The side band frequencies are 1506 kHz and 1494 kHz.
(B) The bandwidth required for amplitude modulation is 6kHz.
(C) The bandwidth required for amplitude modulation is 3 MHz.
(D) The side band frequencies are 1503 kHz and 1497 kHz.

PC0017

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EXERCISE-S

1. Is it necessary for a transmitting antenna to be at the same height as that of the receiving antenna for 'line-of-sight communication? A TV transmitting antenna is 81m tall. How much service area can it cover if the receiving antenna is at the ground level?

PC0001

2. A carrier wave of peak voltage 12 V is used to transmit a message signal. What should be the peak 'voltage of the modulating signal in order to have a modulation index of 75%?

PC0002

3. For an amplitude modulated wave, the maximum amplitude is found to be 10V while the minimum amplitude is found to be 2V. Determine the modulation index μ.

PC0003

4. If the sum of the heights of transmitting and receiving antennas in line of sight of communication is fixed at h, show that the range is maximum when the two antennas have a height h/2 each.

PC0004

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EXERCISE-JM

1. A diode detector is used to detect an amplitude modulated wave of 60% modulation by using a condenser of capacity 250 pico farad in parallel with a load resistance 100 kilo ohm. Find the maximum modulated frequency which could be detected by it. **[JEE Main-2013]**

(1) 10.62kHz (2) 5.31 MHz (3) 5.31 kHz (4) 10.62MHz

PC0020

2. A single of 5 kHz frequency is amplitude modulated on a carrier wave of frequency 2 MHz. The frequencies of the resultant signal is/are - **[JEE Main-2015]**

(1) 2005 kHz, 2000 kHz and 1995 kHz (2) 2000 kHz and 1995 kHz

(3) 2 MHz only (4) 2005 kHz and 1995 kHz

PC0021

3. Choose the correct statement : **[JEE Main-2016]**

(1) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the frequency of the audio signal.

(2) In amplitude modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.

(3) In amplitude modulation the frequency of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.

(4) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.

PC0022

4. In amplitude modulation, sinusoidal carrier frequency used is denoted by ω_c and the signal frequency is denoted by ω_m. The bandwidth ($\Delta\omega_m$) of the signal is such that $\Delta\omega_m << \omega_c$. Which of the following frequencies is not contained in the modulated wave ? **[JEE Main-2017]**

(1) $\omega_m + \omega_c$ (2) $\omega_c - \omega_m$ (3) ω_m (4) ω_c

PC0023

5. A telephonic communication service is working at carrier frequency of 10 GHz. Only 10% of it is utilized for transmission. How many telephonic channels can be transmitted simultaneously if each channel requires a bandwidth of 5 kHz ? **[JEE Main-2018]**

(1) 2×10^4 (2) 2×10^5 (3) 2×10^6 (4) 2×10^3

PC0024

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

6. In a communication system operating at wavelength 800 nm, only one percent of source frequency is available as signal bandwidth. The number of channels accomodated for transmitting TV signals of band width 6 MHz are (Take : velocity of light $c = 3 \times 10^8$m/s, $h = 6.6 \times 10^{-34}$ J-s)

[JEE Main-2019_Jan]

(1) 3.75×10^6 (2) 4.87×10^5 (3) 3.86×10^6 (4) 6.25×10^5

PC0025

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7. The modulation frequency of an AM radio station is 250 kHz, which is 10% of the carrier wave. If another AM station approaches you for license what broadcast frequency will you allot ?

[JEE Main-2019_Jan]

(1) 2750 kHz (2) 2000 kHz (3) 2250 kHz (4) 2900 kHz

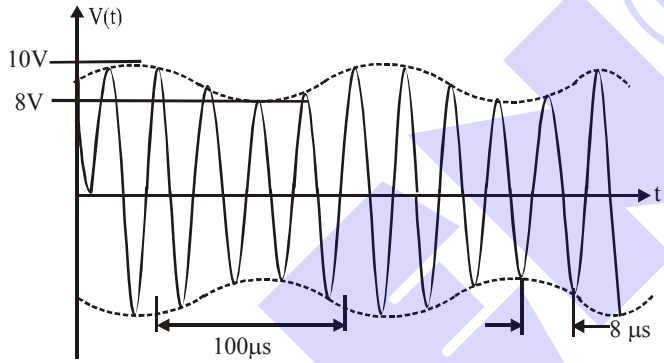
PC0026

8. A TV transmission tower has a height of 140 m and the height of the receiving antenna is 40 m. What is the maximum distance upto which signals can be broadcasted from this tower in LOS (Line of Sight) mode ? (Given : radius of earth = 6.4×10^6 m). **[JEE Main-2019_Jan]**

(1) 80 km (2) 48 km (3) 40 km (4) 65 km

PC0027

9. An amplitude modulated signal is plotted below :- **[JEE Main-2019_Jan]**



Which one of the following best describes the above signal ?

(1) $(9 + \sin(2.5\pi \times 10^5 t))\sin(2\pi \times 10^4 t)$V

(2) $(9 + \sin(4\pi \times 10^4 t))\sin(5\pi \times 10^5 t)$V

(3) $(1 + 9\sin(2\pi \times 10^4 t))\sin(2.5\pi \times 10^5 t)$V

(4) $(9 + \sin(2\pi \times 10^4 t))\sin(2.5\pi \times 10^5 t)$V

PC0028

10. A 100 V carrier wave is made to vary between 160 V and 40 V by a modulating signal. What is the modulation index ? **[JEE Main-2019_Jan]**

(1) 0.6 (2) 0.5 (3) 0.3 (4) 0.4

PC0029

11. The physical sizes of the transmitter and receiver antenna in a communication system are :-

(1) proportional to carrier frequency

(2) inversely proportional to modulation frequency

(3) inversely proportional to carrier frequency

(4) independent of both carrier and modulation frequency **[JEE Main-2019_April]**

PC0030

12. A message signal of frequency 100 MHz and peak voltage 100 V is used to execute amplitude modulation on a carrier wave of frequency 300 GHz and peak voltage 400 V. The modulation index and difference between the two side band frequencies are : **[JEE Main-2019_April]**

(1) 4; 1×10^8 Hz (2) 0.25; 1×10^8 Hz (3) 4; 2×10^8 Hz (4) 0.25; 2×10^8 Hz

PC0031

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13. Given below in the the left column are different modes of communication using the kinds of waves given the right column. **[JEE Main-2019_April]**

A.	Optical Fibre communication	P.	Ultrasound
B.	Radar	Q.	Infrared Light
C.	Sonar	R.	Microwaves
D.	Mobile Phones	S.	Radio Waves

(1) A-S, B-Q, C-R, D-P
(2) A-R, B-P, C-S, D-Q
(3) A-Q, B-S, C-R, D-P
(4) A-Q, B-S, C-P, D-R

PC0032

14. An audio signal $v_m = 20 \sin 2\pi (1500\ t)$ amplitude modulates a carrier

$$v_C = 80 \sin 2\pi (100{,}000\ t).$$

The value of percent modulation is _____. **[JEE Main-2021_Feb]**

PC0033

15. A signal of 0.1 kW is transmitted in a cable. The attenuation of cable is –5 dB per km and cable length is 20 km. The power received at receiver is 10^{-x} W. The value of x is _____.

[Gain in dB = $10 \log_{10}\left(\frac{P_0}{P_i}\right)$] **[JEE Main-2021_Feb]**

PC0034

16. The maximum and minimum amplitude of an amplitude modulated wave is 16V and 8V respectively. The modulation index for this amplitude modulated wave is $x \times 10^{-2}$. The value of x is_____.

[JEE Main-2021_Feb]

PC0035

17. If the highest frequency modulating a carrier is 5 kHz, then the number of AM broadcast stations accommodated in a 90 kHz bandwidth are ..... **[JEE Main-2021_Feb]**

PC0036

18. Two identical antennas mounted on identical towers are separated from each other by a distance of 45km. What should nearly be the minimum height of receiving antenna to receive the signals in line of sight ? (Assume radius of earth is 6400 km) **[JEE Main-2021_March]**

(1) 19.77 m (2) 39.55 m (3) 79.1 m (4) 158.2 m

PC0037

19. Match List–I with List–II. **[JEE Main-2021_March]**

	List–I		**List–II**
(a)	10 km height over earth's surface	(i)	Thermosphere
(b)	70 km height over earth's surface	(ii)	Mesosphere
(c)	180 km height over earth's surface	(iii)	Stratosphere
(d)	270 km height over earth's surface	(iv)	Troposphere

(1) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)
(2) (a)–(i), (b)–(iv), (c)–(iii), (d)–(ii)
(3) (a)–(iii), (b)–(ii), (c)–(i), (d)–(iv)
(4) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)

PC0038

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20. What should be the height of transmitting antenna and the population covered if the television telecast is to cover a radius of 150 km ? The average population density around the tower is 2000/km^2 and the value of $R_e = 6.5 \times 10^6$ m. **[JEE Main-2021_July]**

(1) Height = 1731 m; Population Covered = 1413×10^5

(2) Height = 1241 m; Population Covered = 7×10^5

(3) Height = 1600 m; Population Covered = 2×10^5

(4) Height = 1800 m; Population Covered = 1413×10^8

PC0039

21. The amplitude of upper and lower side bands of A.M. wave where a carrier signal with frequency 11.21 MHz, peak voltage 15 V is amplitude modulated by a 7.7 kHz sine wave of 5V amplitude are $\frac{a}{10}$V and $\frac{b}{10}$V respectively. Then the value of $\frac{a}{b}$ is _____. **[JEE Main-2021_July]**

PC0040

22. A baseband signal of 3.5 MHz frequency is modulated with a carrier signal of 3.5 GHz frequency using amplitude modulation method. What should be the minimum size of antenna required to transmit the modulated signal ? **[JEE Main-2022_June]**

(A) 42.8 m (B) 42.8 mm (C) 21.4 mm (D) 21.4 m

PC0041

23. A signal of 100 THz frequency can be transmitted with maximum efficiency by : **[JEE Main-2022_June]**

(A) Coaxial cable (B) Optical fibre

(C) Twisted pair of copper wires (D) Water **PC0042**

24. A sinusoidal wave $y(t) = 40\sin(10 \times 10^6\ \pi t)$ is amplitude modulated by another sinusoidal wave $x(t) = 20\sin(1000\ \pi t)$. The amplitude of minimum frequency component of modulated signal is : **[JEE Main-2022_June]**

(A) 0.5 (B) 0.25 (C) 20 (D) 10

PC0043

25. Choose the correct statement for amplitude modulation: **[JEE Main-2022_June]**

(A) Amplitude of modulating is varied in accordance with the information signal.

(B) Amplitude of modulated is varied in accordance with the information signal.

(C) Amplitude of carrier signal is varied in accordance with the information signal.

(D) Amplitude of modulated is varied in accordance with the modulating signal.

PC0044

26. We do not transmit low frequency signal to long distances because **[JEE Main-2022_June]**

(a) The size of the antenna should be comparable to signal wavelength which is unreal solution for a signal of longer wavelength.

(b) Effective power radiated by a long wavelength baseband signal would be high.

(c) We want to avoid mixing up signals transmitted by different transmitter simultaneously.

(d) Low frequency signal can be sent to long distances by superimposing with a high frequency wave as well.

Therefore, the most suitable options will be :

(A) All statements are true (B) (a), (b) and (c) are true only

(C) (a), (c) and (d) are true only (D) (b), (c) and (d) are true only **PC0045**

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27. Amplitude modulated wave is represented by $V_{AM} = 10\left[1 + 0.4\cos\left(2\pi \times 10^4 t\right)\right]\cos\left(2\pi \times 10^7 t\right)$. The total bandwidth of the amplitude modulated wave is : **[JEE Main-2022_June]**

(A) 10 kHz (B) 20 MHz (C) 20 kHz (D) 10 MHz

PC0046

28. Match List-I with List-II

	List-I		List-II
A	Television signal	I	03 KHz
B	Radio signal	II	20 KHz
C	High Quality Music	III	02 MHz
D	Human speech	IV	06 MHz

Choose the correct answer from the options given below : **[JEE Main-2022_June]**

(A) A-I, B-II, C-III, D-IV (B) A-IV, B-III, C-I, D-II

(C) A-IV, B-III, C-II, D-I (D) A-I, B-II, C-IV, D-III

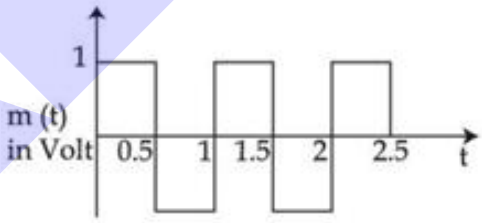
PC0047

29. In AM modulation, a signal is modulated on a carrier wave such that maximum and minimum amplitude are found to be 6V and 2V respectively. The modulation index is **[JEE Main-2022_July]**

(A) 100% (B) 80% (C) 60% (D) 50%

PC0048

30. A square wave of the modulating signal is shown in the figure. The carrier wave is given by C(t) = 5 sin (8 πt) Volt. The modulation index is : **[JEE Main-2022_July]**



(A) 0.2 (B) 0.1 (C) 0.3 (D) 0.4

PC0049

31. A FM Broad cast transmitter, using modulating signal of frequency 20 kHz has a deviation ratio of 10. The Bandwidth required for transmission is : **[JEE Main-2022_July]**

(A) 220 kHz (B) 180 kHz (C) 360 kHz (D) 440 kHz

PC0050

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ANSWER KEY

EXERCISE # O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	A	B	B	B	A	B	B	A
Que.	11	12	13	14						
Ans.	A	C	A,B,D	B,D						

EXERCISE # S

1. No. Service area will be $A = \pi d_7^2 = \frac{22}{7} \times 162 \times 6.4 \times 10^6 = 3258\ km^2$.

2. $\mu = 0.75 = \frac{A_m}{A_c}; A_m = 0.75 \times 12 = 9V$

3. Since the AM wave is given by $(A_c + A_m) \sin \omega_m t) \cos \omega_m t$, the maximum amplitude is $M1 = A_c + A_m$. Hence the modulation index is

$$m = \frac{A_m}{A_c} = \frac{8}{12} = \frac{2}{3}$$

with $M_2 = 0$, clearly, $m = 1$ irrespective of M_1

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	2	3	2	4	2	4	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	4	25	8	33	9	2	1	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	C	B	D	C	C	C	C	D	A
Que.	31									
Ans.	D									

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